Project Synopsis: Wildfire Early Alert – Spatiotemporal Wildfire Risk Scoring (PS-113)

Cover

- **Project title:** Wildfire Early Alert – Spatiotemporal Wildfire Risk Scoring (PS-113)
- **Team name & ID:** Anarchy (Team ID: 100)
- **Institute / Course:** GLA University, Mathura – B.Tech CSE (AI/ML), 2nd year, Mini Project (academic year 2026–27)
- **Version:** v1.0 (first submission)
- **Date:** 9 Oct 2026
- **GitHub repository link:** https://github.com/Himanshu-GenAI/2nd-Year-Mini-Project

Revision history

Version	Date	Author	Change
v0.1	8 Oct 2026	Himanshu Rautela	Initial draft
v1.0	9 Oct 2026	Team	First submission – synopsis completed for PS-113

1. Overview

- **Problem statement:** Uncontrolled forest fires devastate vital ecosystems and rural assets long before ground crews can pinpoint the initial ignition coordinates. Emergency response delays let fires spread across large terrains, and current monitoring lacks the predictive precision to deploy resources before a fire escalates. Authorities need a preventative intelligence system that aggregates real-time environmental metrics to forecast ignition risk before it materialises.
- **Goal:** Build a machine-learning system that fuses 1.88M historical wildfire records with daily weather data (Open-Meteo API) and the Keetch-Byram Drought Index (KBDI) to predict the probability that a new fire is detected within 48 hours in each geographic grid cell (used as a proxy for ignition risk), and present the results as an explainable, interactive risk map for emergency authorities.
- **Non-goals:** Real-time satellite fire detection, fire-spread simulation, automated SMS/mobile alerting, and nationwide production deployment in v1.
- **Value proposition:** Forward-looking (48 h ahead) rather than reactive risk scoring; every score is explained with SHAP drivers; plain-language municipal risk reports from a local language model; and a simple colour-coded heatmap that non-technical officers can read.
- **Problem definition:** For each grid cell and each day t over India, predict the probability that a new fire is detected in that cell during the next 48 hours, using only information available up to day t. A fire counts as new when the cell had no detection in the previous 3 days (days t-2 to t); cells that are already burning are excluded from the primary evaluation, so continuing fires cannot inflate the scores. For comparison we also report an any-fire variant (a detection in the next 48 hours on all cell-days). This is a binary classification problem with extreme class imbalance, because most cell-days have no fire. FIRMS records burning (thermal anomalies), not the moment of ignition, so the model predicts new-fire occurrence, which we use as a proxy for ignition risk.
- **Context in India:** Forest fires recur every year; the fire season is presumed to run from November to June [13]. According to ISFR 2021, as quoted in an FSI proposal, 2.81% of forest cover is extremely fire prone, 7.85% very highly fire prone and 11.61% highly fire prone [13]. National services mainly detect fires that have already started and issue rule-based danger alerts [11][13].
- **Proposed solution:** A pipeline that joins historical fire detections [1] with daily weather [7] and the Keetch-Byram Drought Index [25], trains Random Forest and XGBoost classifiers with strict time-based validation,

explains every high-risk score with SHAP [26], writes a plain-language report with a local language model, and shows the result on an interactive map.

1.1 Methodology and problem-solving approach

- Collect: download FIRMS MODIS C6.1 CSV for India (2015-01-01 to 2025-12-31) [1] and daily weather from Open-Meteo [7].
- Prepare: clean and filter detections (keep type = 0, presumed vegetation fire, and confidence of 30 or more; the standard MODIS archive file (MCD14ML) carries a type field [5], and we check the CSV header in Week 1: if the field is missing, this filter is dropped and a forest or land-cover mask is used instead), build a geographic grid clipped to India's boundary [9] and aggregate to a cell-day table.
- Engineer features: weather lags and rolling sums, KBDI [25], recent fire history in the cell and its neighbours, and calendar features.
- Label and split: the primary label is a new fire (a detection in the next 48 hours in a cell with no detection in the previous 3 days) and the any-fire label is reported as a second variant; train on 2015–2022, validate on 2023, test on 2024–2025, with a 2-day gap between periods to prevent leakage.
- Model: compare climatology and logistic-regression baselines with Random Forest and XGBoost [27]; handle imbalance with class weights or negative subsampling. Both inflate the predicted probabilities, so we apply a prior correction for the subsampling rate and then recalibrate on the untouched 2023 validation year (isotonic or Platt scaling), and check calibration again on the test period.
- Evaluate: PR-AUC [28], recall in the top-k riskiest cells per day, calibration and error analysis by month and state.
- Explain and report: SHAP drivers [26] per high-risk cell, then a short municipal-style report from a local LLM with a number-check guardrail.
- Deliver: FastAPI service and Streamlit/Folium dashboard, tested and documented for demo.

2. Scope and Control

2.1 In-scope

- Data ingestion and cleaning of historical wildfire records; gridding of India into 0.5° × 0.5° cells (about 1,200 cells inside the India boundary; 0.25°, about 4,700 cells, is a stretch goal).
- Daily weather retrieval from Open-Meteo and computation of KBDI and lagged weather features.
- Random Forest / XGBoost classifier for 48-hour new-fire probability with temporal validation and PR-AUC evaluation under extreme class imbalance.
- SHAP explainability, local-LLM plain-language risk reports, FastAPI backend and Streamlit/Folium dashboard with spatiotemporal heatmaps.

2.2 Out-of-scope

- Live satellite or sensor integration, fire-spread modelling, evacuation planning, mobile apps, and paid cloud deployment in v1.

2.3 Assumptions

- Historical wildfire data comes from NASA FIRMS (MODIS Collection 6.1 active-fire product [4][5][6]), exported as CSV for India only, 1 Jan 2015 to 31 Dec 2025 [1][2][3], and is openly available for academic use.
- Open-Meteo free tier provides sufficient daily weather coverage for the chosen study region.
- Grid and data volume: at 0.5° the cell-day table has about 1,200 cells × 4,018 days ≈ 4.8 million rows (2015–2025); at 0.25° it would have about 4,700 cells ≈ 19 million rows. Cell counts are estimates and are confirmed in Week 1 after the India boundary is applied.
- Study region is India only (state boundaries from the DataMeet community maps [9]) and the study period is 2015–2025 (11 years); if compute is limited, a state-level or year-range subset will be fixed in Week 1.

2.4 Constraints

- 8-week timeline, 3-member student team, laptop/free-tier compute (e.g. Colab/Kaggle), and a locally run language model of limited size.

2.5 Dependencies

- Open-Meteo API, NASA FIRMS MODIS C6.1 CSV export (registration with FIRMS by email verification for archive downloads, and a FIRMS MAP_KEY if the API is used [1][3]; if the bulk download fails or times out, the request is split into yearly pieces through the web form, or the country yearly summary files [2] or the API are used instead), Python ML stack (pandas, scikit-learn, XGBoost, SHAP), FastAPI, Streamlit, Folium, and a small open-source language model (3B to 8B parameters, quantised) run locally with Ollama; the exact model is chosen after speed tests in Week 5, before the Week 6 report work.

2.6 Acceptance criteria and sign-off

- GIVEN a region and a date WHEN the user requests a risk map THEN the dashboard shows a colour-coded heatmap of 48-hour new-fire probability within 5 seconds.
- GIVEN a high-risk grid cell WHEN the user opens its details THEN the top SHAP drivers and a plain-language explanation are displayed.
- GIVEN the held-out test period WHEN the model is evaluated THEN PR-AUC is reported, is at least 2× the base rate and exceeds the baseline models, with no train/test time overlap.
- **Sign-off:** Team lead and all members approve the demo and test report; all P1 bugs closed (severity levels are defined in Section 13.2). No external mentor is assigned.

Sign-off table

Stakeholder	Role	Decision area
Himanshu Rautela	Team Lead	Scope, release readiness, final acceptance
Dhruv Bhardwaj	Data & Backend Engineer	Data pipeline and API sign-off
Ashutosh Chaudhary	Dashboard, QA & Docs	Dashboard, test report and documentation sign-off

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Himanshu	Himanshu	—	Team
Data pipeline & features	Dhruv	Himanshu	—	Team
Modelling & validation	Himanshu	Himanshu	—	Team
Explainability & LLM reports	Himanshu, Ashutosh	Himanshu	—	Team
API & integration	Dhruv	Himanshu	—	Team
Dashboard	Ashutosh	Himanshu	—	Team
Testing & documentation	Ashutosh, Dhruv	Himanshu	—	Team
Release & demo	Team	Himanshu	—	Dept
Submission and evaluation	Team	Himanshu	—	Technical Trainer (AIML), GLA University

- Stakeholders: the three team members and the Technical Trainer (AIML) at GLA University, who receives and evaluates the synopsis. No external mentor is assigned.
- Because the team lead is accountable for every area, each member has a named backup who reviews their pull requests and can take over: Dhruv backs up Himanshu, Ashutosh backs up Dhruv, and Himanshu backs up Ashutosh.

4. Team and Roles

Roles confirmed by the team.

Member	Role	Responsibilities	Key skills	Contact
Himanshu Rautela	Team Lead & ML Engineer	Scope, model training/validation, SHAP, LLM reports, reviews	Python, ML, scikit-learn/XGBoost	himanshu.rautela_cs.aiml25@gla.ac.in
Dhruv Bhardwaj	Data & Backend Engineer	Ingestion, Open-Meteo, KBDI/feature pipeline, FastAPI	Python, pandas, FastAPI	dhruv.bhardwaj_cs.aiml25@gla.ac.in
Ashutosh Chaudhary	Frontend, QA & Docs	Streamlit/Folium dashboard, test plan, documentation, slides	Streamlit, Folium, testing	ashutosh.chaudhary_cs.aiml25@gla.ac.in

5. Week-wise Plan and Assignments

Assumes the 8-week window runs 12 Oct – 6 Dec 2026; shift dates if your official calendar differs.

Week	Dates	Milestones	Himanshu (Lead/ML)	Dhruv (Data/Backend)	Ashutosh (UI/QA)	Deliverables	Status
1	12–18 Oct	Requirements & data audit	Finalise scope, region, labels	Download and profile datasets	Literature review, wireframes	Synopsis, data audit note	Planned
2	19–25 Oct	Data pipeline & grid	Grid definition, target label (48 h)	Ingestion, cleaning, Open-Meteo fetch	Test plan v0, UI layout	Clean dataset, grid spec	Planned
3	26 Oct–1 Nov	Feature engineering	Temporal split design, baseline model	KBDI + lag/rolling features	Streamlit + Folium skeleton	Feature store v1, baseline	Planned
4	2–8 Nov	Model training	RF/XGBoost, imbalance handling	FastAPI scaffold, /risk endpoint	Heatmap on mock data	Model v1, PR-AUC report	Planned
5	9–15 Nov	Explainability & integration	SHAP analysis, model tuning, LLM speed test and model choice	Serve model via API	Connect dashboard to API	Feature A demo (heatmap)	Planned
6	16–22 Nov	LLM reports	Local LLM prompts, report quality	/explain and /report endpoints	SHAP charts, report panel	Feature B demo	Planned
7	23–29 Nov	Hardening & testing	Final model, error analysis	Bug fixes, API tests	UI/E2E tests, regression	Test report	Planned
8	30 Nov–6 Dec	Release & final deck	Final report, sign-off	README, run instructions	User manual, slides	v1.0, slides, demo	Planned

6. Users and UX

6.1 Personas

- **District Forest Officer:** Needs a quick view of which areas are most likely to ignite in the next 48 hours so crews can be pre-positioned; values clarity and speed.
- **Municipal Risk Planner:** Needs a plain-language summary and the reasons behind a risk score to justify resource decisions; values explainability.

6.2 Top user journeys

- Officer: Open dashboard → Select region and date → View colour-coded heatmap → Click high-risk cell → See score and drivers. KPI: ≤ 4 steps, map load ≤ 5 s.
- Planner: Open dashboard → Select region → Generate risk report → Read/export summary. KPI: report generated ≤ 15 s.

6.3 User stories

- As a forest officer, I want a 48-hour risk heatmap so that I can pre-position crews before a fire starts. GIVEN I select a region and date WHEN I press “Show risk” THEN I see a heatmap with a colour legend.
- As a planner, I want to know why a cell is high risk so that I can trust the score. GIVEN I click a cell WHEN details open THEN I see the top contributing factors and a short text explanation.

6.4 Accessibility & localization

- Colour palette that remains readable for colour-blind users (with numeric score labels); language: English; RTL not required.

6.5 Real-world applications and use cases

- Pre-positioning fire watchers and crews during the November–June season, based on the next-48-hour risk map.
- Prioritising patrol routes, fire-line maintenance and awareness drives in the highest-risk blocks.
- Complementing existing alerts: a forest department can overlay predicted risk on near-real-time detections from the FSI alert system [11].
- District and municipal planning: plain-language reports that explain why an area is risky, useful for briefings and public advisories.
- Research and teaching: a reproducible, open-data benchmark for fire occurrence prediction in India.

6.6 Target customers and market segment

- **Primary users:** State forest departments and forest-fire nodal officers (who already receive FSI alerts [13]), and district disaster-management authorities.
- **Secondary users:** Protected-area managers (national parks, tiger reserves), municipal and rural local bodies, environmental NGOs and researchers.
- **Possible later segments:** Infrastructure operators near forests and insurers, similar to the commercial customers of existing wildfire-risk software [18][19].
- **Market segment:** GovTech and climate-tech (disaster-risk analytics), with a business-to-government focus. v1 is an academic prototype with no revenue model; we did not find a verified market-size figure, so none is claimed.

7. Market and Competitors

7.1 Existing solutions and competitor table

Competitor	Product	Target users	Key features	Pricing / access	Strengths	Weaknesses	Our differentiator
FSI Forest Fire Alert System (FAST 3.0) [11][12][13]	Government alert service, India	State forest departments and registered public users	Near-real-time SMS/email alerts from MODIS and SNPP-VIIRS detections; large-fire alerts; pre-fire alerts from a customised Fire Weather Index	Government service; registration required	National reach (about 1.6 lakh SMS subscribers per an FSI proposal); official forest mask and data	Alerts correspond to satellite overpasses and FSI advises strategic use only; pre-fire alerts are rule-based, not learned probabilities	Learned 48 h probability per grid cell, with per-prediction explanations and plain-language reports

Competitor	Product	Target users	Key features	Pricing / access	Strengths	Weaknesses	Our differentiator
NASA FIRMS [1][16]	Satellite active-fire data portal	Researchers, agencies and managers in 160+ countries	Near-real-time MODIS and VIIRS hotspots through web map, email alerts and downloads	Free	Global coverage, trusted source, open archive	Reports fires that are already burning; no forecast	Uses FIRMS history as training data to forecast fire occurrence
Global Forest Watch Fires [15]	Web monitoring platform	NGOs, land managers, public	Near-real-time fire alerts (MODIS 1 km, VIIRS 375 m), area subscriptions, fire reports	Free	Easy maps, alert subscriptions, historical comparisons	Alert-based; no fire-risk probability	Cell-level risk forecast with drivers, built for India
Canadian Fire Weather Index System [17]	Rule-based fire danger rating	Fire agencies	Six components calculated from daily temperature, humidity, wind and 24 h rainfall	Method is published and free	Well established, simple, widely used	Fixed equations based on weather only; not learned from local fire history; no probability output	Learns from local fire history with KBDI and weather, with calibrated probabilities
Technosylva Wildfire Analyst [18][19]	Commercial SaaS	Utilities and fire agencies	Risk forecasting and fire-spread simulation using fuel, weather, terrain and asset data	Commercial subscription (pricing not public)	Mature, high-resolution, used by many agencies	Cost and detailed fuel, terrain and asset inputs needed	Free, open-data, explainable option suitable for pilots in India

7.2 Market research summary

- India already has a national alert service. FSI sends near-real-time alerts, large-fire alerts and pre-fire alerts, and pre-fire alerts come from a customised Fire Weather Index [11][13].
- FSI's own FAQ says its alerts cover only the satellite overpasses in a day and should be used for strategic purposes, which leaves room for forward-looking tools [11].
- FSI's Centre of Excellence proposal plans to adopt AI/ML methods, which shows demand for this direction [13].
- The research review by Jain et al. covers ML for fire occurrence, susceptibility and risk, notes opportunities for newer methods, and asks for freely available data [23]. Huot et al. built a feature-rich dataset with weather and a drought index and compared logistic regression with random forest, which supports our feature and baseline choices [24].
- India-specific studies point the same way. A 2024 random-forest study for India reported rainfall and vegetation as the most important drivers of fire occurrence [21]; a next-day risk model for Uttarakhand describes MODIS/VIIRS detections as reactive [22]; and Chandel et al. found that FIRMS detected fewer than 20% of burned areas in Central India because fires there are small and crop-residue and forest fires occur side by side [20].
- Commercial systems exist but are enterprise SaaS products aimed at utilities and agencies [18][19].
- Open data is available: the FIRMS archive [1] and Open-Meteo historical weather [7], which is free for non-commercial use with rate limits [8].

7.3 Gap and positioning

- **Gap:** To the best of our research (October 2026), we did not find a free, open-data tool for India that gives a calibrated 48-hour fire-occurrence probability per grid cell together with a per-prediction explanation and a plain-language report.
- **Differentiators:** (1) forward-looking probability instead of detection only; (2) SHAP explanation for every high-risk cell; (3) plain-language reports from a local model; (4) open, reproducible pipeline; (5) honest evaluation with PR-AUC and time-based validation.

- **Measurable delta:** PR-AUC above simple baselines on a time-held-out test set, plus an explanation for every flagged cell.

8. Objectives and Success Metrics

- **O1 Model quality:** On the 2024–2025 test period, by 20 Nov 2026, PR-AUC is at least 2× the base rate (the PR-AUC of a random classifier, equal to the share of positive cell-days) and above the climatology and logistic-regression baselines, and recall@top-10% (the share of new-fire cells on a given day that fall in the 10% highest-scored cells, averaged over fire-season days) is at least 40%. Both label variants are reported. The targets are set now and reviewed once baseline results exist (end of Week 3). KPI: PR-AUC, recall@top-10%.
- **O2 Validation integrity:** Four leakage checks, all automated as tests: (a) every feature for day t uses only data up to day t; (b) scalers, imputers, climatology baselines and per-cell statistics (including the mean annual rainfall used for KBDI) are fitted on training years only; (c) label windows never cross a split boundary (2-day gap between train, validation and test); (d) no cell-day appears in more than one split. KPI: all four checks pass.
- **O3 API performance:** p95 latency of /risk for one region-day ≤ 2 s by 29 Nov 2026. KPI: p95 ms.
- **O4 Explainability:** SHAP top-5 drivers available for 100% of cells above the high-risk threshold. KPI: coverage %.
- **O5 Report generation:** Local-LLM risk report generated in ≤ 15 s per region. KPI: seconds.

Expected outcomes

- A working end-to-end prototype for India: data pipeline, trained model, API and dashboard.
- A documented model comparison showing PR-AUC against baselines on the 2024–2025 test period.
- An explainable risk map with the top drivers for each high-risk cell.
- A plain-language report generator for regional summaries.
- A reproducible repository with README, tests and a final report, and practical experience for the team in ML on imbalanced spatiotemporal data.

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Data ingestion & gridding	Load/clean 1.88M fire records; map to grid cells; daily weather from Open-Meteo	Must	Dataset, Open-Meteo	GIVEN raw data WHEN pipeline runs THEN a clean cell-day table is produced
Feature engineering (KBDI)	KBDI, lag and rolling weather features	Must	Ingestion	GIVEN cell-day data WHEN features run THEN KBDI and lags are added with no future data
Fire labels (new-fire and any-fire)	Primary label new_fire_next_48h and secondary label fire_next_48h built from the FIRMS detections	Must	Ingestion, grid	GIVEN the cell-day table WHEN labels are built THEN new_fire_next_48h is 1 only if a detection occurs in the next 48 hours and none occurred in the previous 3 days, cells already burning are excluded from the primary evaluation, and fire_next_48h is 1 for any detection in the next 48 hours
Risk model	RF/XGBoost 48-hour new-fire probability with temporal validation	Must	Features, labels	GIVEN the test period WHEN evaluated THEN PR-AUC is reported for both label variants, is at least 2x the base rate on the new-fire label and beats both baselines
SHAP explainability	Per-cell driver attribution	Must	Model	GIVEN a flagged cell WHEN opened THEN top drivers shown
LLM risk reports	Local LLM turns scores + drivers into municipal text	Should	SHAP, LLM runtime	GIVEN a region WHEN report requested THEN text is returned ≤ 15 s
FastAPI backend	Endpoints for risk, explanation and report	Must	Model	GIVEN request WHEN called THEN valid JSON/GeoJSON returned

Feature	Description	Priority	Dependencies	Acceptance criteria
Streamlit/Folium dashboard	Interactive spatiotemporal heatmap	Must	API	GIVEN region/date WHEN selected THEN colour-coded heatmap renders

Detailed flows and edge cases will be kept in design docs in the repository.

10. Architecture

10.1 High-level

- **Clients:** Streamlit web dashboard with Folium maps
- **Services:** FastAPI service (risk, explain, report); offline training pipeline
- **Data stores:** Parquet files / SQLite for features and predictions; saved model artefacts
- **Integrations:** Open-Meteo API; local LLM runtime

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes
/api/risk	GET	—	48 h risk for grid cells	region, date	200 {cells[{id, lat, lon, score}]}	200, 400
/api/explain/{cell_id}	GET	—	SHAP drivers for a cell	cell_id, date	200 {drivers[]}	200, 404
/api/report	POST	—	Plain-language region report	region, date	200 {report_text}	200, 400, 500
/api/health	GET	—	Service health check	—	200 {status}	200

10.3 Config and secrets

- .env for local settings (API base URL, model path); git-ignored. Open-Meteo needs no key for non-commercial use; any future keys are kept out of the repository.

10.4 Technologies, AI/ML models and tools

Layer	Technology	Purpose
Data processing	Python, pandas, PyArrow (Parquet) [37], GeoPandas [32]	Cleaning, gridding, cell-day table, storage
Data sources	NASA FIRMS MODIS C6.1, India area [1][2][3]; Open-Meteo historical weather at India grid points [7]; DataMeet India state boundaries [9]	Fire detections and daily weather
Drought feature	Keetch-Byram Drought Index [25]	Cumulative dryness feature per cell
ML models	Climatology and logistic regression baselines; Random Forest and XGBoost [27][29][30]	48-hour fire-occurrence probability
Evaluation	PR-AUC [28], recall at top-k, Brier score, reliability (calibration) plot	Metrics suitable for extreme imbalance
Explainability	SHAP [26][31]	Top drivers per cell
Language model	Small open model run locally with Ollama [36]	Plain-language risk reports
API	FastAPI [33]	Risk, explanation and report endpoints
Dashboard	Streamlit [34] and Folium [35]	Interactive risk map
Quality	pytest, ruff, GitHub Actions	Tests, linting, CI

10.5 Deployment strategy

- **Phase 1 (v1, mini project):** local deployment. A batch script computes predictions; the FastAPI service (Uvicorn) and the Streamlit dashboard run on a team laptop for the demo. Models and data are stored as versioned files.
- **Reproducibility:** pinned dependencies and a documented run order; Docker packaging is a stretch goal.

- **Phase 2 (future):** host the API and dashboard on an institute server or a free-tier cloud VM, refresh weather and predictions with a scheduled nightly job, and use template-based reports when no GPU is available for the language model.
- **Phase 3 (future):** connect to alert channels such as SMS or email for forest departments; this is out of scope for v1.

11. Data Design

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
GridCell	cell_id	String	No	—	System	PK
GridCell	lat/lon bounds	Float	No	Valid coordinates	System	Cell extent
FireRecord	latitude, longitude	Float	No	Valid coordinates in India	NASA FIRMS (MODIS C6.1)	Used to assign each detection to a grid cell
FireRecord	acq_date	Date	No	2015-01-01 to 2025-12-31	NASA FIRMS (MODIS C6.1)	Detection date (UTC); used for labels
FireRecord	acq_time	Int (HHMM)	No	0000–2359	NASA FIRMS (MODIS C6.1)	Detection time (UTC)
FireRecord	confidence	Int	No	0–100	NASA FIRMS (MODIS C6.1)	Filter: keep 30 or more
FireRecord	type	Int	Yes	0–3	NASA FIRMS (MODIS C6.1)	Keep 0 (presumed vegetation fire); see Section 1.1
FireRecord	frp	Float	Yes	0 or more	NASA FIRMS (MODIS C6.1)	Fire radiative power
WeatherDaily	temp, humidity, wind, precip	Float	Yes	Physical ranges	Open-Meteo	Per cell per day
FeatureRow	kbdi	Float	No	0–800 (KBDI scale, mm-based variants possible)	Computed	Drought index
FeatureRow	new_fire_next_48h	Int	No	0 / 1	Derived	Primary label: detection in next 48 h and none in previous 3 days
FeatureRow	fire_next_48h	Int	No	0 / 1	Derived	Secondary label: any detection in next 48 h
Prediction	risk_score	Float	No	0–1	Model	Probability

11.2 Schemas and migrations

- ER diagram is in Section 12; schemas are versioned in the repository and data is regenerated by the pipeline scripts.

11.3 Privacy, retention, backup/DR

- No personal data – only public fire and weather records. Raw data retained for the project duration. Code on GitHub; datasets and model artefacts backed up to cloud storage. RTO: 24 h, RPO: 1 week.

11.4 Grid and data volume

- Grid: regular 0.5° × 0.5° cells clipped to the India boundary [9], about 1,200 cells; the grid size is a configuration value so 0.25° (about 4,700 cells) can be tried if weather downloads finish in time.

- Volume: 2015-01-01 to 2025-12-31 is 4,018 days, so the cell-day table has about 4.8 million rows at 0.5° (about 19 million at 0.25°), stored as Parquet partitioned by year.

11.5 KBDI baseline setup

- KBDI is calculated one day at a time for each cell from daily maximum temperature and rainfall, the cell's mean annual rainfall (computed from the 2015–2022 training years only) and a fixed soil-moisture capacity: the index runs from 0 (saturated) to 800 (maximum drought) in hundredths of an inch, which assumes an 8-inch capacity. Each cell starts from a spin-up year (2014) before the study period [25].

12. Technical Workflow Diagram

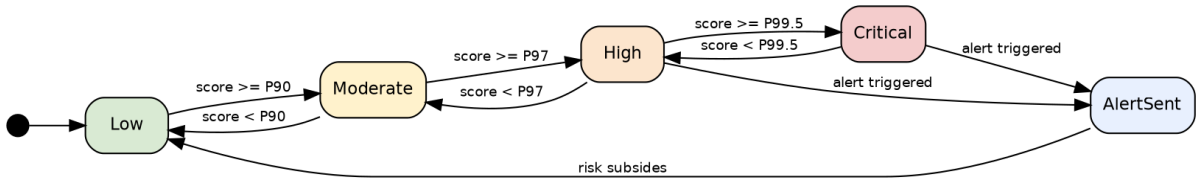


Figure 12.1 – State Transition Diagram (risk tier of a grid cell; tiers are percentile-based, Pk = k-th percentile of validation-year scores)

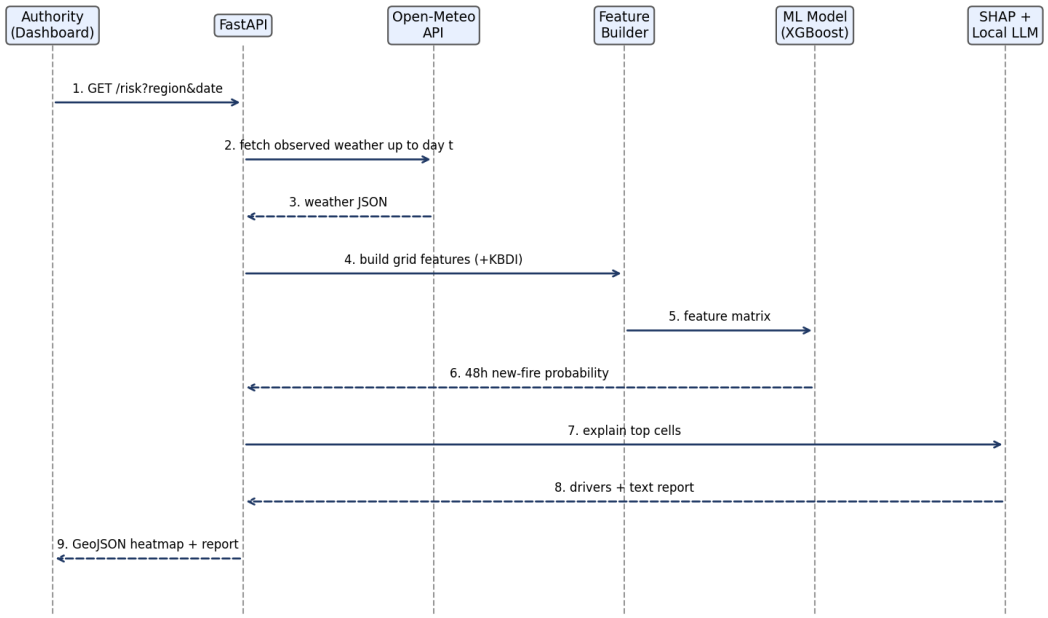


Figure 12.2 – Sequence Diagram (risk map request using observed weather up to day t)

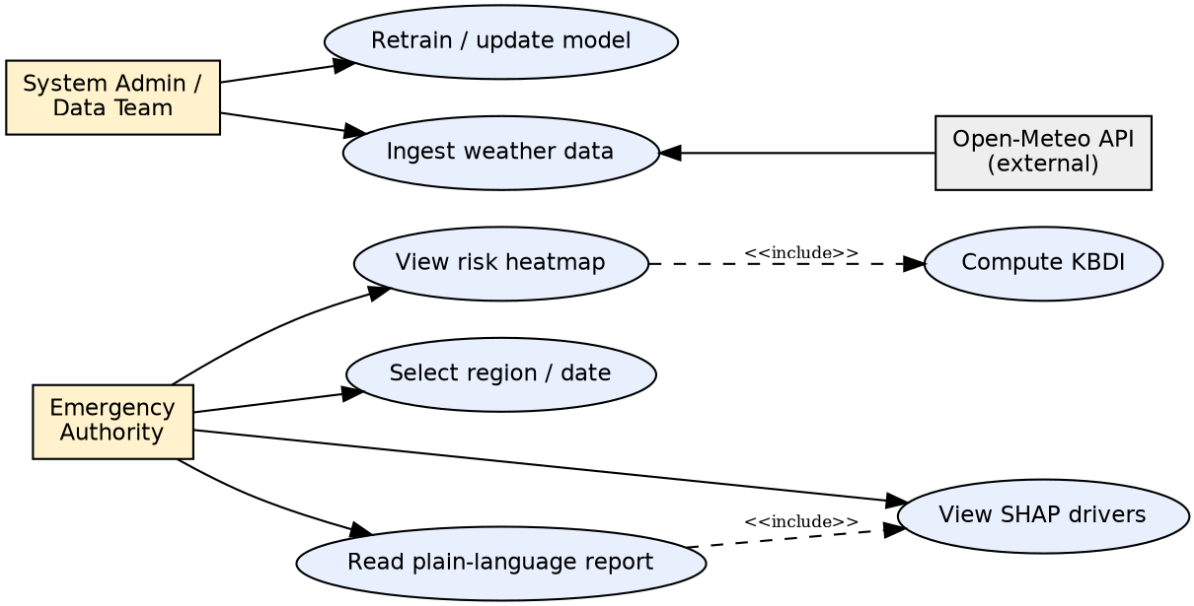


Figure 12.3 – Use Case Diagram

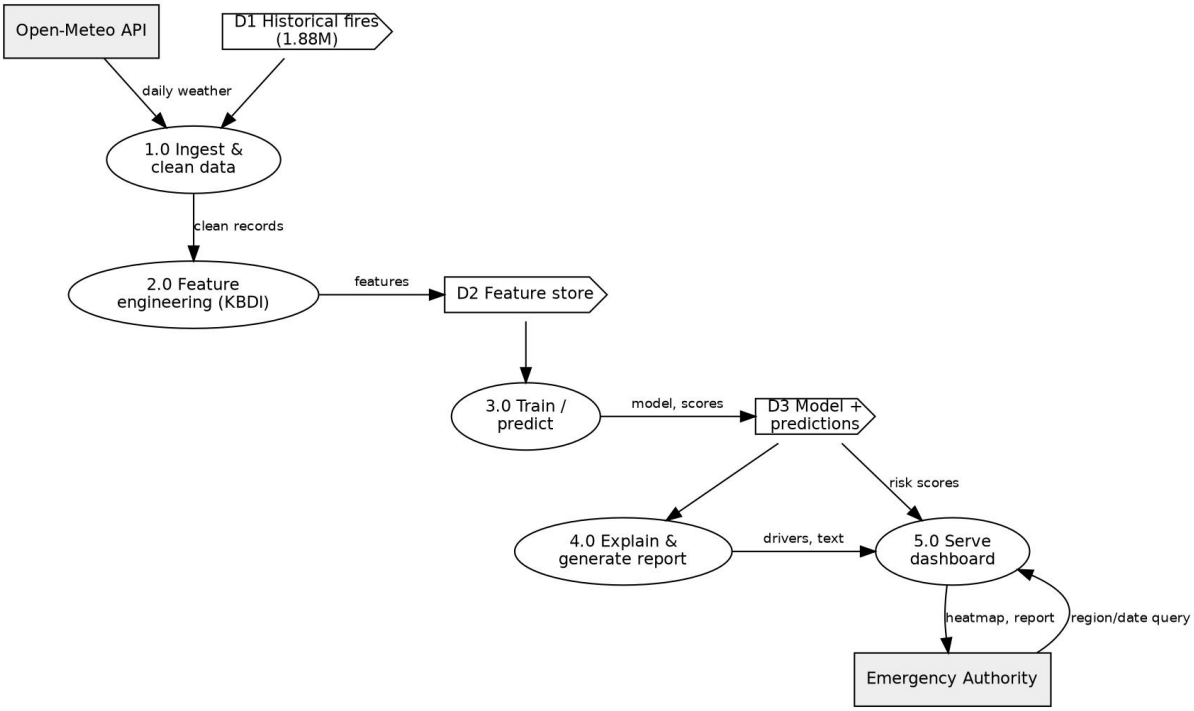


Figure 12.4 – Data Flow Diagram

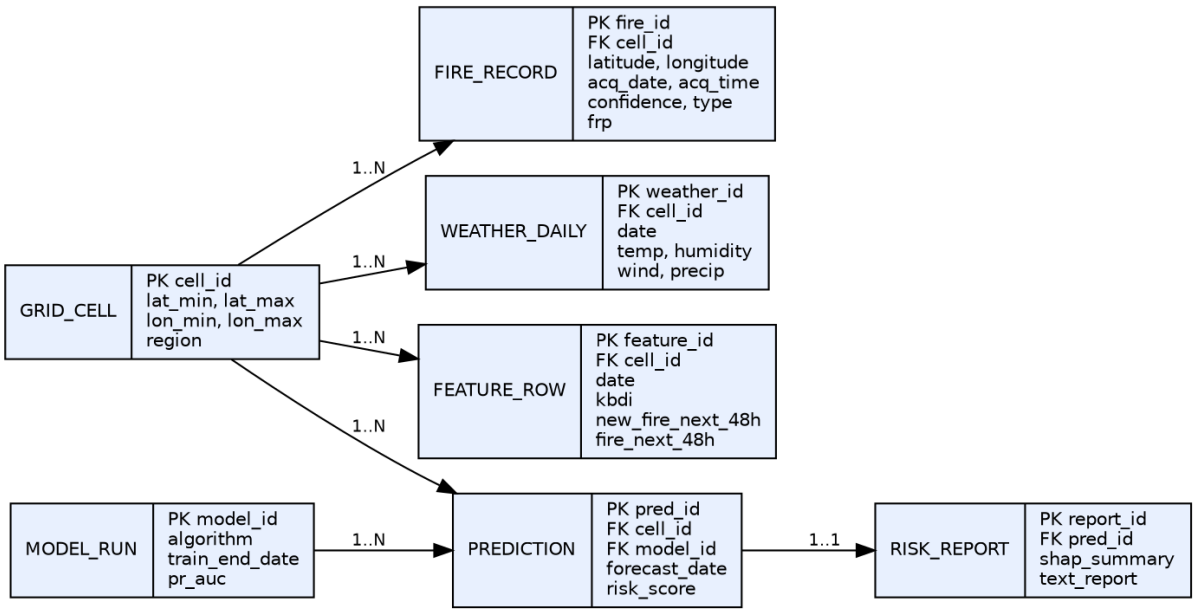


Figure 12.5 – ER Diagram

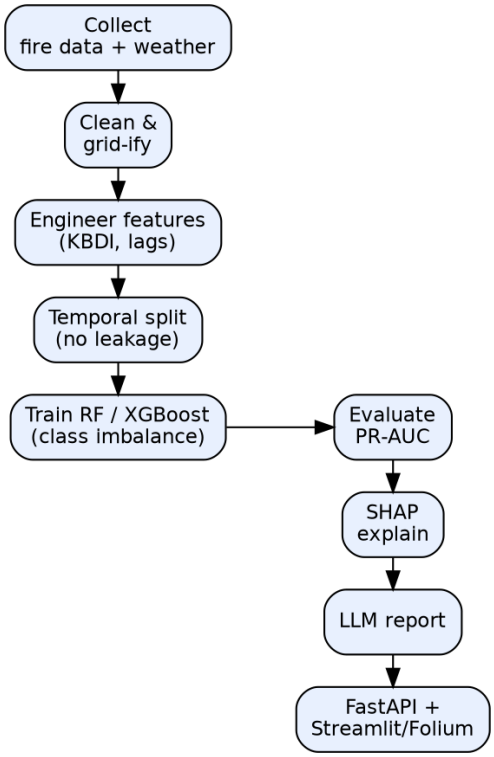


Figure 12.6 – Technical Workflow Diagram (data to dashboard)

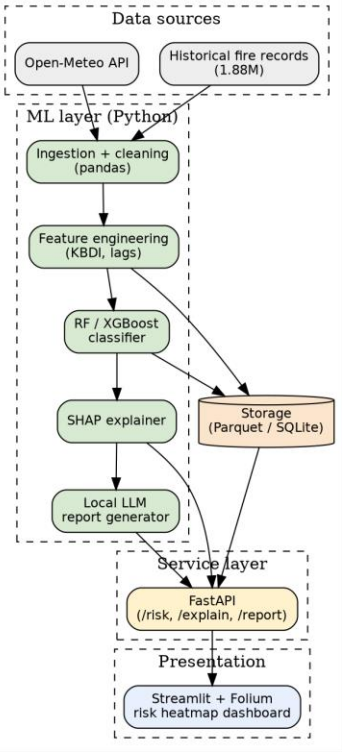


Figure 12.7 – Work Architecture Diagram

13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Latency	p95 /risk	≤ 2000 ms	API timing logs
Model quality	PR-AUC and recall@top-10% (test period)	PR-AUC ≥ 2× base rate and above baselines; recall@top-10% ≥ 40%	Evaluation notebook
Report time	LLM seconds/report	≤ 15 s	Timing logs
Calibration	Brier score and reliability plot (validation and test periods)	Brier score below the climatology baseline; plot included in the final report	Evaluation notebook
Reproducibility	Pipeline re-run	Same results with fixed seed	Manual check

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Feature code (KBDI, lags)	Unit	pytest	Dhruv	70%	No P1/P2 defects
Model validation	Leakage checks (a)–(d) from objective O2 and temporal split tests	pytest, scikit-learn, notebooks	Himanshu	All four checks	All checks pass; PR-AUC beats baseline
API	Integration	pytest + FastAPI TestClient	Dhruv	All endpoints	All critical pass
Dashboard	Manual / E2E	Streamlit, manual scripts	Ashutosh	Core journeys	Pass rate ≥ 95%

Bug severity levels used in the exit criteria and sign-off:

Level	Definition	Rule
P1 Critical	Crash, wrong or misleading risk output, data or label leakage, or anything that blocks the demo	Must be fixed before sign-off

Level	Definition	Rule
P2 Major	A main feature (map, drivers, report, API endpoint) fails or gives wrong explanation text, but a workaround exists	Fixed before release, or accepted in writing by the team lead
P3 Minor	Cosmetic or small usability issue with no effect on results	Fix if time allows
P4 Trivial	Typo, wording or enhancement idea	Logged for future scope

13.3 Environments

- Local development → Colab/Kaggle for heavy training → local demo environment; Git branches for risky changes.

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
API	Request flooding	Denial of service	Medium	Low	Rate limiting, input validation	Dhruv
Model files	Tampering with artefacts	Tampering	Medium	Low	Checksums, access control on repo	Himanshu
Config/secrets	Leak via repository	Information disclosure	Medium	Medium	.env git-ignored, secret scanning	Himanshu
LLM reports	Prompt injection or steered output	Information disclosure / Elevation of privilege	Low	Low	Model runs locally with no tools, no secrets in the prompt and no user free text; input is only structured fields from our own model; number-check guardrail and template fallback	Ashutosh

14.2 AuthN/AuthZ

- AuthN: none for the academic demo; optional API-key header can be added. Roles: viewer (dashboard) and admin (retrain/update) enforced at API layer if enabled.

14.3 Audit and logging

- Log API requests, model runs and errors; 30-day retention locally.

14.4 Compliance

- Academic project; follow university policy; respect dataset and Open-Meteo licence/attribution terms; language model runs locally so no data leaves the machine.

15. Delivery and Operations

15.1 Release plan

- v1.0 demo in Week 8 (week of 30 Nov 2026); features gated in the dashboard until stable.

15.2 CI/CD and rollback

- CI: GitHub Actions – lint → pytest → build; releases tagged in Git; rollback by checking out the previous tag.

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 latency	> 2500 ms	Dhruv	“API Latency” runbook
Error rate	> 2%	Dhruv	“Error Spike” runbook

15.4 Runbooks

- **API Latency:** check data-loading code → cache predictions → reduce grid size.
- **Error Spike:** inspect logs → roll back to last tag → note the incident.

15.5 Communication plan

- Regular team stand-ups and a weekly internal status review; demo at mid-point (Week 4) and Week 8.

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Extreme class imbalance misleads metrics	High	High	16	Use PR-AUC, class weights/resampling, baseline comparison	Himanshu	Open
Temporal data leakage	Medium	High	12	Strict time-based splits with a 2-day gap; automated leakage checks (O2)	Himanshu	Open
1.88M records strain laptop memory	Medium	Medium	9	Parquet, chunked processing, region/year subset	Dhruv	Open
Open-Meteo rate limits / gaps	Medium	Medium	9	Batch requests, cache locally, fill gaps	Dhruv	Open
FIRMS bulk download fails or is incomplete	Low	High	8	Request yearly pieces through the Archive Download form [1]; fall back to country yearly summaries [2] or the API with a MAP_KEY [3]; keep a backup copy of every downloaded file; verify row counts per year	Dhruv	Open
Local LLM too slow or weak	Medium	Medium	9	Small quantised model, template fallback text	Himanshu	Open
Schedule slip	Medium	High	12	Scope freeze in Week 3, weekly demos	Himanshu	Open
Single-point dependency on the team lead (accountable for all areas)	Medium	High	12	Named backup for each member, Dhruv as deputy lead, decisions logged in docs/decisions.md, every pull request reviewed by a second person	Himanshu	Open

17. Research and Evaluation

17.1 Market and literature research

- Existing services reviewed: FSI FAST 3.0 [11][12][13][14], NASA FIRMS [1][16], Global Forest Watch Fires [15], the Canadian FWI System [17] and Technosylva [18][19]. The findings and gap are in Section 7.
- Research reviewed: India-focused fire studies [20][21][22], ML in wildfire science [23], the Next Day Wildfire Spread dataset [24], the KBDI paper [25], SHAP [26], XGBoost [27] and PR-curve evaluation under imbalance [28].
- Data and technology sources are listed with links in Section 18.

17.2 Evaluation plan

- Evaluation deliverables for the final report: a table of PR-AUC and recall@top-10% for both label variants and both baselines, a reliability (calibration) plot with Brier score for the validation and test periods, and a SHAP summary plot.
- Track the KPIs in Section 8 weekly; compare against baselines on a future held-out period; review SHAP drivers against known fire-weather behaviour (dry, hot, windy days with high KBDI should raise risk).

17.3 Scalability

- Data volume: Parquet files partitioned by year, chunked processing, and Dask or Spark if the data outgrows one machine; process state by state if needed.
- Resolution: move from 0.5° to 0.25° and finer cells; add VIIRS 375 m detections, which FSI and Global Forest Watch also use [11][15].
- Geography: the pipeline is configuration-driven, so other regions or countries can be added using FIRMS downloads [1].
- Serving: predictions are precomputed and the API is stateless, so it can be replicated behind a load balancer.

17.4 Future scope

- Use weather forecasts as inputs so predictions look further ahead than the latest observations, and cross-check rainfall against IMD gridded data (0.25° rainfall) [10].
- Add land cover, vegetation (NDVI), terrain, population and road-proximity data, and overlay a forest-cover mask, as FSI does for its alerts [11], to separate forest fires from crop-residue burning.
- Try spatiotemporal deep-learning models, ensembles and uncertainty estimates.
- Add SMS, WhatsApp or email alerts and reports in Hindi and other regional languages.
- Create a feedback loop with forest departments to validate predictions against ground reports, and overlay predictions on FSI alerts.

17.5 Limitations

- Satellite detections capture fires only at overpass times and can miss small or short-lived fires [11]. In Central India, FIRMS detected fewer than 20% of burned areas in one Landsat-based comparison, because fires there are small [20]; the model therefore learns from detected fires, not from every fire.
- The label is new-fire occurrence (a detection within 48 hours in a cell with no detection in the previous 3 days), not confirmed ignition; many Indian detections are agricultural burning, and forest and crop-residue fires occur close together [20].
- Weather uncertainty, no fuel-type data in v1, and results limited to the chosen grid and period.

18. Appendices

18.1 Glossary

KBDI – Keetch-Byram Drought Index; PR-AUC – area under the precision-recall curve; SHAP – SHapley Additive exPlanations; RF – Random Forest; XGBoost – gradient-boosted trees; LLM – large language model; FWI – Fire Weather Index; FIRMS – Fire Information for Resource Management System; FSI – Forest Survey of India; ISFR – India State of Forest Report; MODIS/VIIRS – satellite sensors used for fire detection; p95 – 95th percentile.

18.2 References and sources

Study area: India only. Online sources were accessed in October 2026.

Datasets and data services (India)

[1]NASA FIRMS, “Archive download: MODIS Collection 6.1 active fire data (CSV), area of interest: India, 1 Jan. 2015 to 31 Dec. 2025,” NASA Earth Science Data and Information System (ESDIS). [Online]. Available: https://firms.modaps.eosdis.nasa.gov/download/ (accessed 9 Oct. 2026).

[2]NASA FIRMS, “Country yearly summary of fire detections (select India).” [Online]. Available: https://firms.modaps.eosdis.nasa.gov/country/. Accessed: Oct. 2026.

[3]NASA FIRMS, “FIRMS API: download fire detections by area or by country code (India = IND).” [Online]. Available: https://firms.modaps.eosdis.nasa.gov/api/. Accessed: Oct. 2026.

[4]L. Giglio, W. Schroeder, and C. O. Justice, “The Collection 6 MODIS active fire detection algorithm and fire products,” *Remote Sens. Environ.*, vol. 178, pp. 31–41, 2016, doi: 10.1016/j.rse.2016.02.054.

[5]NASA, “MODIS Collection 6 and Collection 6.1 active fire product user’s guide,” Version 1.0, NASA Earthdata. [Online]. Available: https://earthdata.nasa.gov/s3fs-public/2023-09/MODIS_C6_C6.1_Fire_User_Guide_1.0.pdf. Accessed: Oct. 2026.

[6]NASA Earthdata, “Fire Information for Resource Management System (FIRMS): products, citation and acknowledgement.” [Online]. Available: https://www.earthdata.nasa.gov/learn/find-data/near-real-time/firms. Accessed: Oct. 2026.

[7]Open-Meteo, “Historical Weather API documentation” (queried at grid-cell centres inside India). [Online]. Available: https://open-meteo.com/en/docs/historical-weather-api. Accessed: Oct. 2026.

[8]Open-Meteo, “Weather data for multiple locations at once,” Open-Meteo Blog. [Online]. Available: https://openmeteo.substack.com/p/weather-data-for-multiple-locations. Accessed: Oct. 2026.

[9]DataMeet Community Maps Project, “India – State boundaries” and “India – District boundaries” (CC BY 2.5 India); repository: https://github.com/datameet/maps. [Online]. Available: https://projects.datameet.org/maps/states and https://projects.datameet.org/maps/districts/. Accessed: Oct. 2026.

[10]India Meteorological Department (IMD), Pune, gridded daily rainfall (0.25°) and temperature (1.0°) data, accessed through the IMDLIB Python library (optional cross-check source). [Online]. Available: https://imdlib.readthedocs.io/en/latest/. Accessed: Oct. 2026.

Existing solutions and market research (India first)

[11]Forest Survey of India, “FSI Forest Fire Alerts System” and “Frequently asked questions.” [Online]. Available: https://fsiforestfire.gov.in/ and https://fsiforestfire.gov.in/document/faq.pdf. Accessed: Oct. 2026.

[12]Forest Survey of India, “FSI Forest Fire Alert System (FAST 3.0),” Technical Information Series, vol. 1, no. 2. [Online]. Available: https://fsiforestfire.gov.in/libraries/assets/documents/technical-information-series-v1-n2.pdf. Accessed: Oct. 2026.

[13]Forest Survey of India, “Proposal for establishment of Centre of Excellence in Forest Survey of India” (Section 4.4, Forest fire monitoring), Ministry of Environment, Forest and Climate Change, Dehradun, India. [Online]. Available: https://nationalcampa.nic.in/dashboard/schemesPDF/66aca1d62814a.pdf. Accessed: Oct. 2026.

[14]“India to get real time forest fire alert system,” Down To Earth. [Online]. Available: https://www.downtoearth.org.in/news/forests/india-to-get-real-time-forest-fire-alert-system-62857. Accessed: Oct. 2026.

[15]S. Minnemeyer, S. Sargent, and M. Weisse, “Fighting fires with satellites: VIIRS fire data now available on Global Forest Watch,” Global Forest Watch Blog. [Online]. Available: https://www.globalforestwatch.org/blog/data/fighting-fires-with-satellites-viirs-fire-data-now-available-on-global-forest-watch. Accessed: Oct. 2026.

[16]NASA, “NASA’s Fire Information for Resource Management System (FIRMS): Near real-time global fire monitoring using data from MODIS and VIIRS,” EARSeL Forest Fires SIG Workshop, NASA Technical Reports Server, Doc. ID 20190032007. [Online]. Available: https://ntrs.nasa.gov/citations/20190032007. Accessed: Oct. 2026.

[17]Natural Resources Canada, “Canada’s Fire Weather Index System.” [Online]. Available: https://natural-resources.canada.ca/forests-forestry/wildland-fires/canada-fire-weather-index-system; and Canadian Wildland Fire Information System, “Canadian Forest Fire Weather Index (FWI) System.” [Online]. Available: https://cwfis.cfs.nrcan.gc.ca/background/summary/fwi. Accessed: Oct. 2026.

[18]Technosylva, “Wildfire Analyst: FireSim.” [Online]. Available: https://technosylva.com/products/wildfire-analyst/firesim/. Accessed: Oct. 2026.

[19]“Why Technosylva runs half a billion simulations daily,” Latitude Media. [Online]. Available: https://www.latitudemedia.com/news/why-technosylva-runs-half-a-billion-simulations-daily/. Accessed: Oct. 2026.

Research papers (India-focused studies first)

[20]A. Chandel, W. Sarwat, A. Najah, S. Dhanagare, and M. Agarwala, “Evaluating methods to map burned area at 30-meter resolution in forests and agricultural areas of Central India,” *Front. For. Glob. Change*, vol. 5, 2022, Art. no. 933807, doi: 10.3389/ffgc.2022.933807.

[21]"New model can better predict areas vulnerable to forest fires in India," *Eos*, vol. 105, 2024, doi: 10.1029/2024EO240307. [Online]. Available: https://eos.org/articles/new-model-can-better-predict-areas-vulnerable-to-forest-fires-in-india. Accessed: Oct. 2026.

[22]"Next-day forest fire risk prediction using machine learning and multimodal satellite data" (study area: Uttarakhand, India), Odisha University of Technology and Research, Bhubaneswar, India, Sciforum paper 27921. [Online]. Available: https://sciforum.net/paper/view/27921. Accessed: Oct. 2026.

[23]P. Jain, S. C. P. Coogan, S. G. Subramanian, M. Crowley, S. Taylor, and M. D. Flannigan, "A review of machine learning applications in wildfire science and management," *Environ. Rev.*, vol. 28, no. 4, pp. 478–505, 2020, doi: 10.1139/er-2020-0019. [Online]. Available: https://arxiv.org/abs/2003.00646

[24]F. Huot, R. L. Hu, N. Goyal, T. Sankar, M. Ihme, and Y.-F. Chen, "Next Day Wildfire Spread: A machine learning dataset to predict wildfire spreading from remote-sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022. [Online]. Available: https://arxiv.org/abs/2112.02447

[25]J. J. Keetch and G. M. Byram, "A drought index for forest fire control," USDA Forest Service, Southeastern Forest Experiment Station, Asheville, NC, USA, Res. Paper SE-38, 1968. [Online]. Available: https://research.fs.usda.gov/treesearch/40

[26]S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017. [Online]. Available: https://arxiv.org/abs/1705.07874

[27]T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, 2016. [Online]. Available: https://arxiv.org/abs/1603.02754

[28]T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015, doi: 10.1371/journal.pone.0118432.

Technologies and tools

[29]scikit-learn developers, "scikit-learn: Machine learning in Python." [Online]. Available: https://scikit-learn.org/

[30]XGBoost developers, "XGBoost documentation." [Online]. Available: https://xgboost.readthedocs.io/

[31]S. Lundberg et al., "SHAP documentation." [Online]. Available: https://shap.readthedocs.io/

[32]GeoPandas developers, "GeoPandas." [Online]. Available: https://geopandas.org/

[33]S. Ramírez, "FastAPI." [Online]. Available: https://fastapi.tiangolo.com/

[34]Streamlit Inc., "Streamlit documentation." [Online]. Available: https://docs.streamlit.io/

[35]Folium developers, "Folium documentation." [Online]. Available: https://python-visualization.github.io/folium/

[36]Ollama, "Ollama." [Online]. Available: https://ollama.com/

[37]Apache Software Foundation, "Apache Arrow: PyArrow documentation" and "Apache Parquet." [Online]. Available: https://arrow.apache.org/docs/python/ and https://parquet.apache.org/